



MANIPAL UNIVERSITY
JAIPUR

Department of Mathematics and Statistics

Assignment # 3, MAS2001

Course Cover: Theory of Estimation: Maximum Likelihood and method moment estimation, Sufficient statistics, Bayesian estimation, Confidence intervals for means.

Q(A). SECTION A	
A1	The criteria of good estimators are.....
A2	Write any two methods of estimation.....
A3	If M.L.E, exists, it is the most in the class of such estimators.
A4	Estimate the parameter p of the binomial distribution by the method of moments.
A5	Let T_1 and T_2 be statistics with expectations $E(T_1)=q_1 + q_2$ and $E(T_2)=q_1 - q_2$. Find unbiased estimators of q_1 and q_2 .
A6	Under what circumstances can the normal distribution be used to find confidence limits of the populations mean?
A7	The intervalis known as the 95% confidence interval for m , and the 95% confidence limits are
A8	Under Bayesian approach, the parameter of a distribution is a.....
A9	Bayes estimator is always a function of..... statistics
A10	Give an example of an estimator (i) Which is consistent but not unbiased (ii) which is unbiased but not consistent.
Q (B). SECTION B	
B1	Let $x_1, x_2, \dots, x_n$ be a random sample from $N(\mu, \sigma^2)$ population. Find sufficient estimators for μ , and σ^2 .
B2	Let $x_1, x_2, \dots, x_n$ be a random sample from a uniform population on $[0, q]$. Find sufficient estimators for q .
B3	In random sampling from normal population $N(\mu, \sigma^2)$, find the maximum likelihood estimators for (i) μ when σ^2 is known (ii) σ^2 when μ is known
B4	A quality control expert is required to estimate the mean thickness of aluminum sheets used in the production of airframes. A sample of 100 sheets reveals a mean of 0.048 inches with a standard deviation of 0.01 inches. Construct the 99 per cent confidence interval.
B5	Show that in estimating for the parameter in the Poisson distribution with general term $\frac{e^{-\mu}}{x!} \mu^x$, $\bar{x}$ is sufficient for μ .
Q (C). SECTION-C	

C1	Let $x_1 = 0.7, x_2 = 1.5, x_3 = 0.3, x_4 = 0.5, x_5 = 3.4$ be a sample of size 5 from a distribution having pdf $f(x, \theta) = \theta e^{-\theta x}, x > 0, \theta > 0$. Find the MLE of θ
C2	Ashu tossed a biased coin 50 times and got head 20 times, while Boby tossed it 90 times and got 40 heads. Find the maximum likelihood estimate of the probability of getting head when the coin is tossed.
C3	A random sample of size 10 was drawn from a normal population with an unknown mean and a variance of 44.1 (inch) ² . If the observations are (in inches): 65, 71, 80, 76, 78, 82, 68, 72, 65 and 81, obtain the 95% confidence interval for the population mean.
C4	Find Bays estimator of the parameter p of a binomial distribution with x successes out of n given that the prior distribution of p is a beta distribution with parameters α and β .
C5	The mean and variance of a random sample of 64 observations were computed as 160 and 100 respectively. (i) compute the 95% confidence limits for population mean. (ii) If the investigator wants to be 95% confident that error in estimate of population mean should not exceed ± 1.4 , how many additional observations are required?
Q (D). SECTION-D	
D1	For the Poisson distribution: $p(x)=P(X=x) = \frac{1}{2} \frac{e^{-m_1} m_1^x}{x!} + \frac{1}{2} \frac{e^{-m_2} m_2^x}{x!}; x = 0, 1, 2, \dots$ show that the estimates for m_1 and m_2 by the methods of moments are: $\mu'_1 \pm \sqrt{\mu'_2 - \mu'_1 - \mu'^2_1}$
D2	$X_1, X_2, \text{ and } X_3$ is a random sample of size 3 from a population with mean value μ and variance σ^2 . T_1, T_2, T_3 are the estimators used to estimate mean value μ , where $T_1 = X_1 + X_2 - X_3, T_2 = 2X_1 + 3X_3 - 4X_2$ and $T_3 = \frac{1}{3}(\lambda X_1 + X_2 + X_3)$. (i) Are T_1 and T_2 unbiased estimators? (ii) Find the value of λ such that T_3 is unbiased estimators for μ . (iii) With this value of λ is T_3 a consistent estimator? (iv) Which is the best estimator?
D3	Find Bayesian estimator of the single parameter λ of the Poisson distribution when it is known that the prior distribution of λ is a gamma distribution.
D4	In a sample survey of 1000 housewives in a city, 23 per cent preferred a particular brand of pressure cooker. Find 99 per cent confidence limits for the percentage of all housewives in the city preferring that brand of cooker.
D5	A manufacturing concern wants to estimate the average amount of purchase of its product in a month by the customers. If the standard deviation is Rs. 10 find the sample size if the maximum error is not to exceed Rs. 3 with a probability of 0.99.